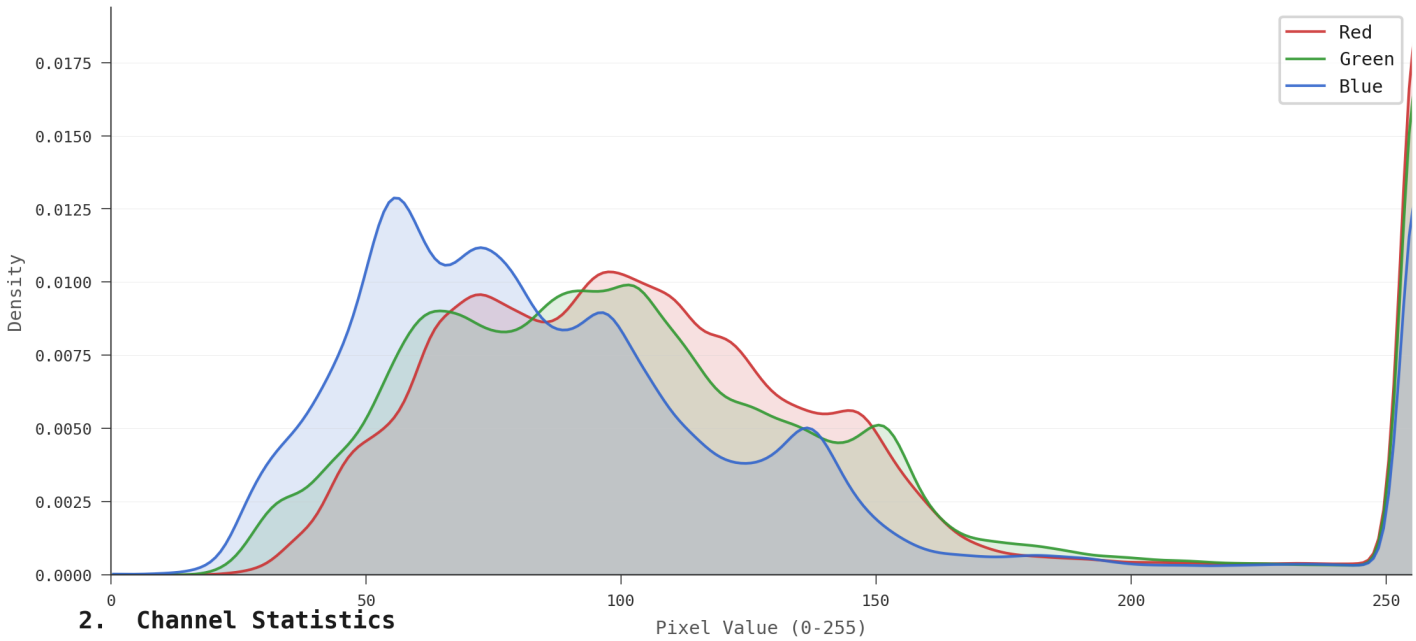
Reference Data Sheet: KODIM24

Subject: Bavarian Painted House

768x512 | 393,216 px
24-bit RGB | sRGB | PNG (lossless)

1. RGB Channel Distribution



2. Channel Statistics

Channel	Mean	Std Dev	Variance	Kurtosis	Skewness	Min	Max
Red	113.4662	53.3450	2845.6931	1.3742	1.3082	20	255
Green	109.5455	54.5518	2975.8954	1.1848	1.2183	10	255
Blue	94.2684	53.1052	2820.1571	2.3358	1.5677	0	255

3. Inter-Channel Correlation

	R	G	B
R	1.0000	0.9761	0.9389
G	0.9761	1.0000	0.9649
B	0.9389	0.9649	1.0000

Pairwise Pearson Correlation Coefficients

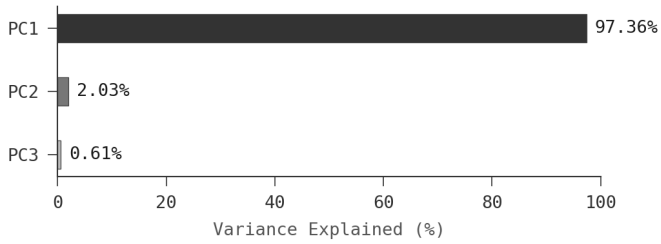
r(R-G)	=	**0.976092**
r(R-B)	=	**0.938867**
r(G-B)	=	**0.964925**
Avg |r|	=	**0.959961**

4. Covariance Matrix S

$$
\begin{array}{c} \\ R \\ G \\ B \end{array}
\begin{array}{c}
\begin{array}{ccc} R & G & B \end{array} \\
\begin{bmatrix}
\mathbf{2845.7003} & 2840.4995 & 2659.7188 \\
2840.4995 & \mathbf{2975.9029} & 2795.3752 \\
2659.7188 & 2795.3752 & \mathbf{2820.1643}
\end{bmatrix}
\end{array}
$$

Reference Data Sheet: KODIM24

5. Eigendecomposition



Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.5728	**-0.5911**	-0.5678	8413.5006
PC2	-0.6349	-0.1181	0.7635	175.8511
PC3	0.5184	-0.7979	0.3077	52.4158

6. Derived Metrics

Condition Number

160.51

(l1/l3) PC1/PC2: 47.84 | PC2/PC3: 3.35

Blue Channel Independence

3.8%

Fraction of blue variance not captured by PC1

PC1 Dominant Channel

Green

Loading: -0.5911 | Balanced

Classification: Near-Degenerate (PC1 = 97.36%)

7. Spatial Autocorrelation (lag-1)

Channel	Horizontal	Vertical	Average
Red	0.942699	0.932641	0.937670
Green	0.936831	0.926359	0.931595
Blue	0.925966	0.914864	0.920415

8. Average Local Variance (3x3)

Channel	Value
Red	253.8905
Green	292.1783
Blue	312.6529

9. Redundancy Profile

Classification: Near-Degenerate | Condition Number: 160.51 | PC1: 97.36%

Correlation: R–G pair dominates at r = 0.976092 | G–B: 0.964925 | R–B: 0.938867

Blue Channel: 3.8% independent | 96.2% captured by PC1

Spatial: Avg autocorrelation: 0.929893 | Avg local variance: 286.2406

Eigenvector: Balanced | R (-0.5728), G (-0.5911), B (-0.5678) near-equal loading

References

[1] Eastman Kodak Company. Kodak Publication No. PCD-042, 1992.

[2] Baetzel, J. (2026). "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."

[3] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.

[4] Giorgianni, E.J. and Madden, T.E. Digital Color Management. Addison-Wesley, 1998.

Blue Channel Independence = (1 - (blue_loading_PC1^2 x l1 / Var(B))) x 100

Data Source: Statistics derived from Baetzel (2026), "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."